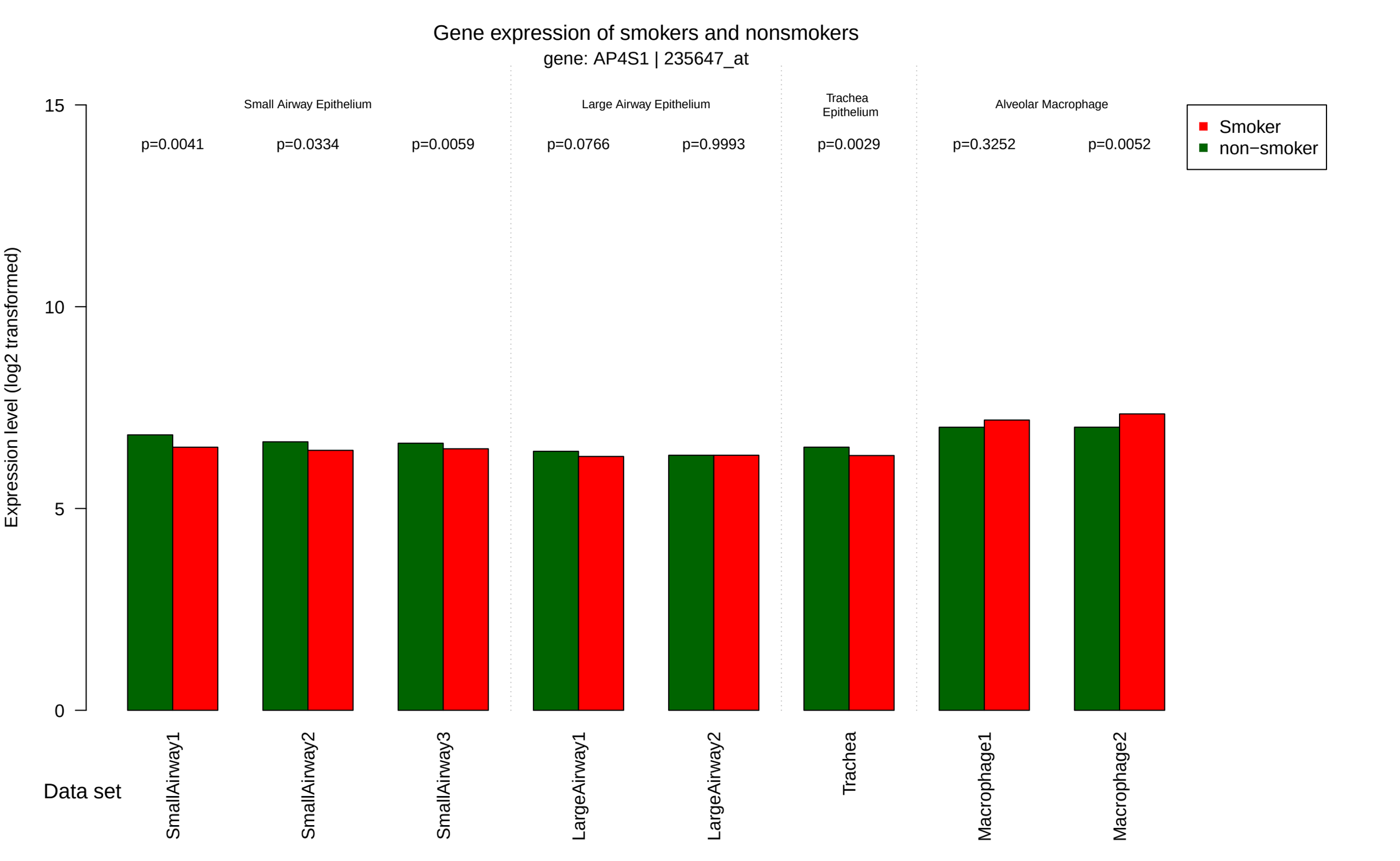
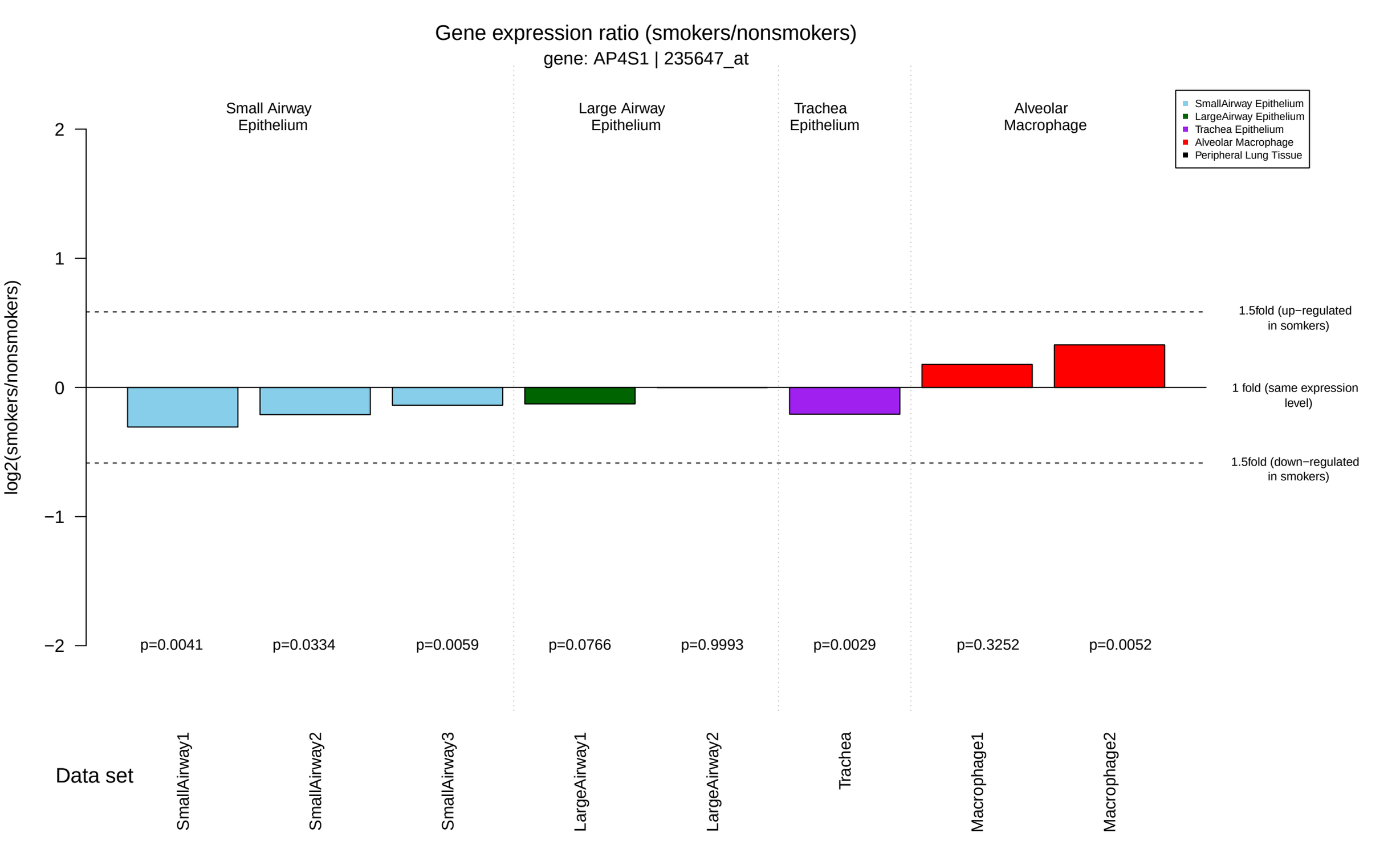
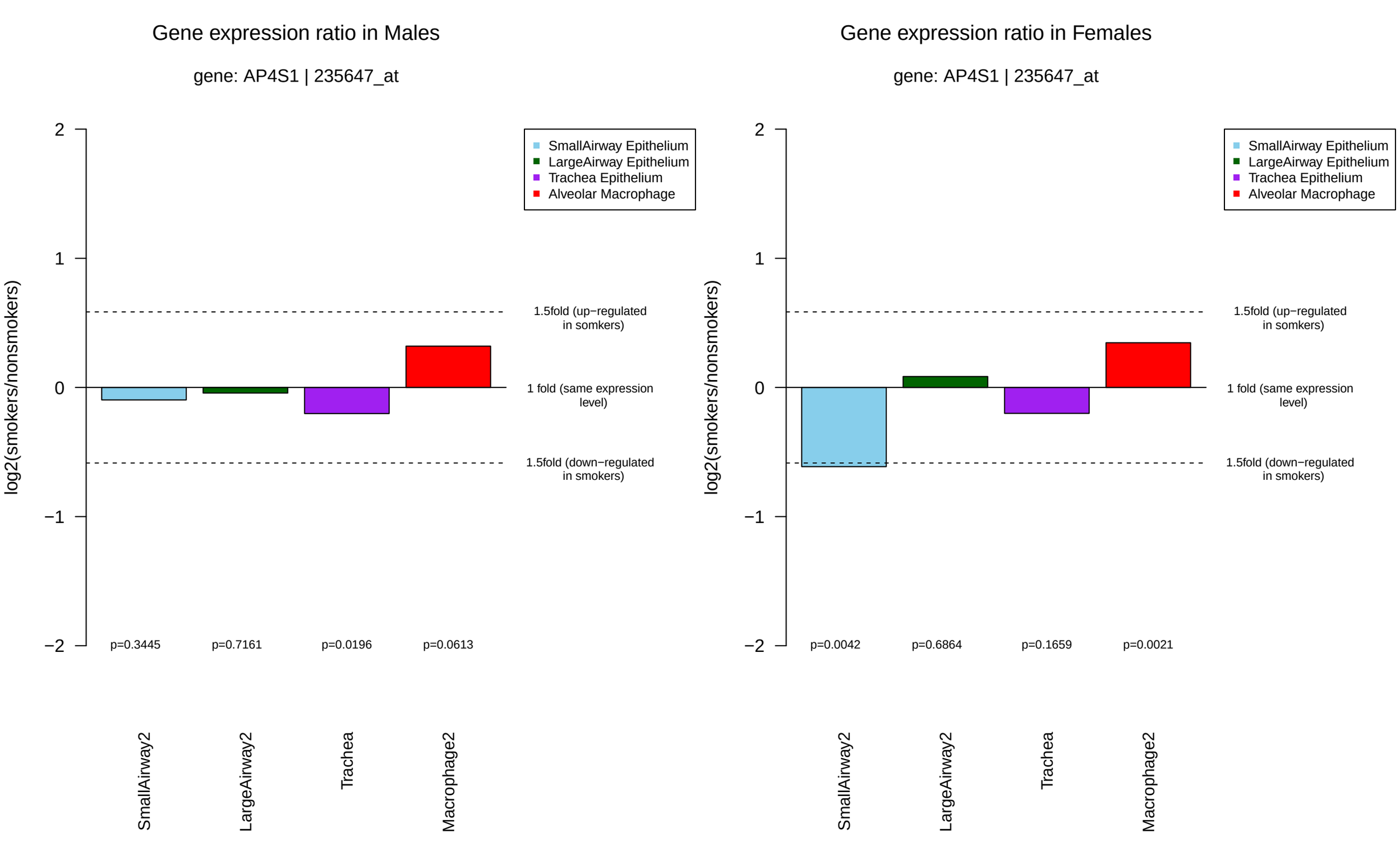
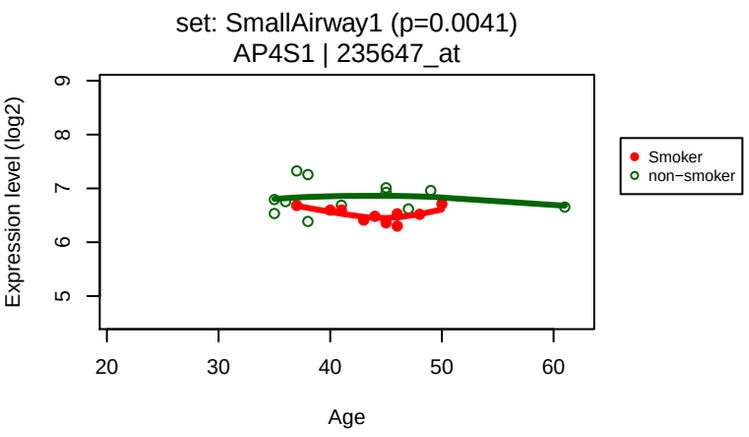
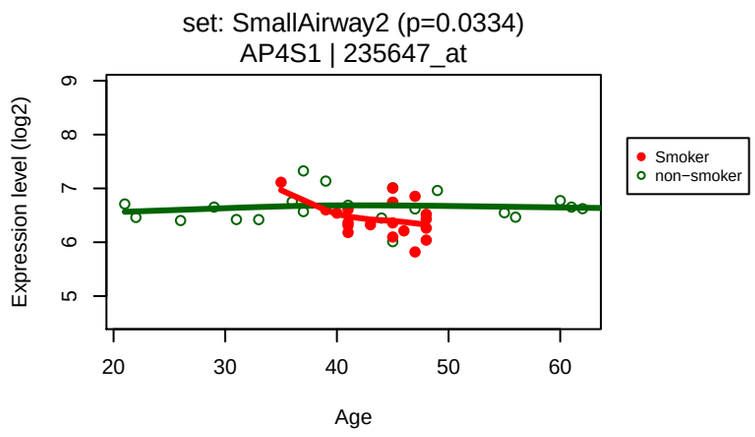
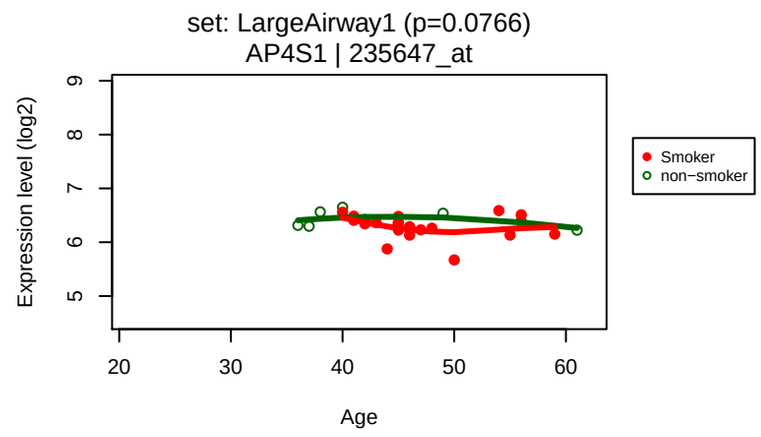

Gene expression of smokers and nonsmokers
gene: AP4S1 | 235647_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0041
p=0.0334
p=0.0059
p=0.0766
p=0.9993
p=0.0029
p=0.3252
p=0.0052
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

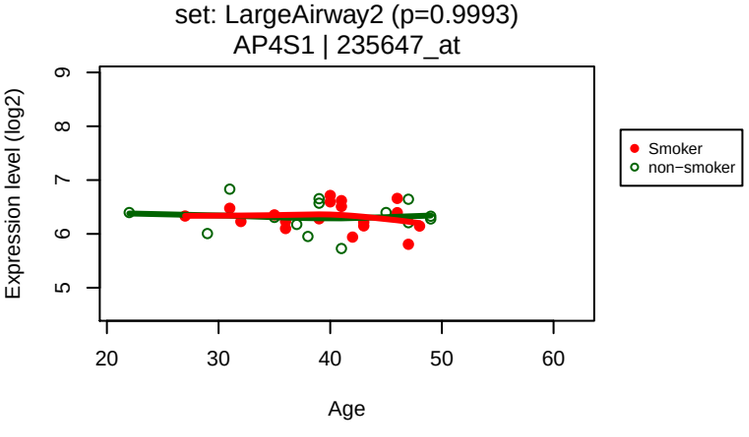

Gene expression ratio (smokers/nonsmokers)
gene: AP4S1 | 235647_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0041
p=0.0334
p=0.0059
p=0.0766
p=0.9993
p=0.0029
p=0.3252
p=0.0052
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

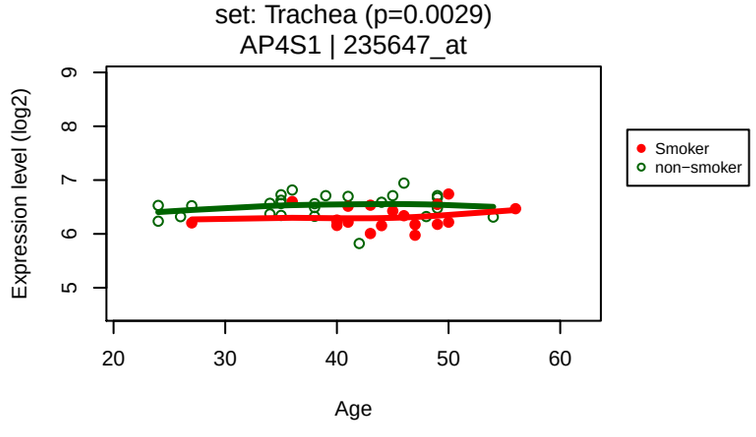

Gene expression ratio in Males
gene: AP4S1 | 235647_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3445
p=0.7161
p=0.0196
p=0.0613
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AP4S1 | 235647_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0042
p=0.6864
p=0.1659
p=0.0021
SmallAirway2
LargeAirway2
Trachea
Macrophage2

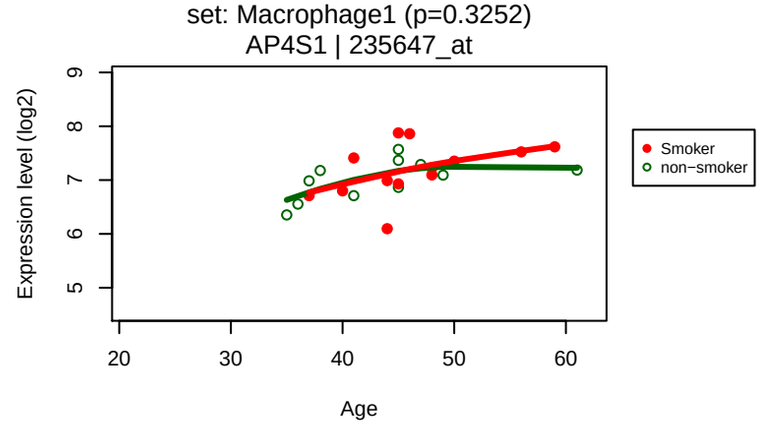

set: SmallAirway1 (p=0.0041)
AP4S1 | 235647_at
Expression level (log2)
Age
Smoker
non−smoker

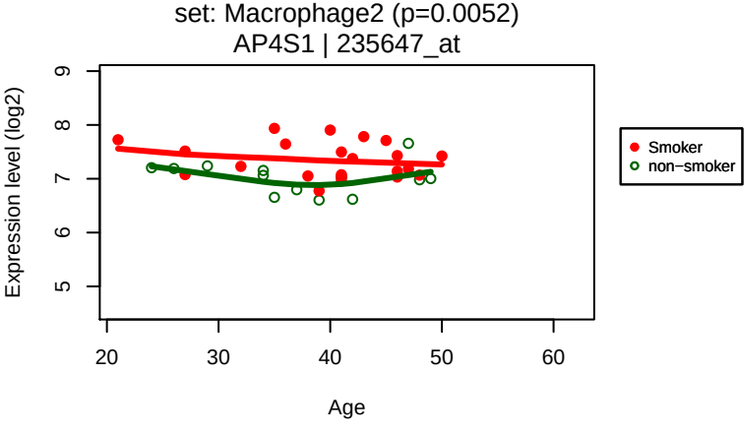

set: SmallAirway2 (p=0.0334)
AP4S1 | 235647_at
Expression level (log2)
Age
Smoker
non−smoker

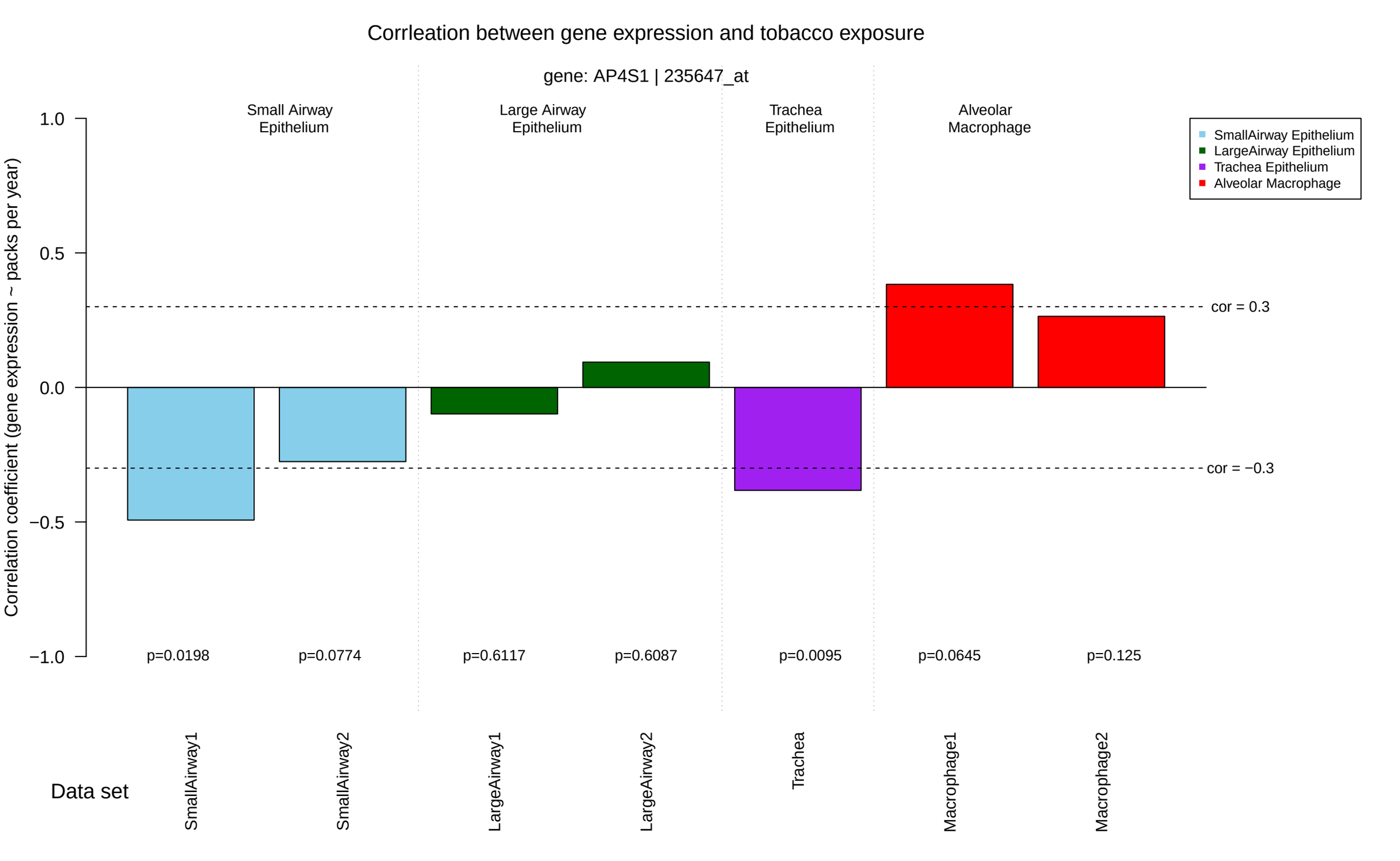

set: LargeAirway1 (p=0.0766)
AP4S1 | 235647_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9993)
AP4S1 | 235647_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0029)
AP4S1 | 235647_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3252)
AP4S1 | 235647_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0052)
AP4S1 | 235647_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AP4S1 | 235647_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0198
p=0.0774
p=0.6117
p=0.6087
p=0.0095
p=0.0645
p=0.125
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2